



A

MRI OF THE BRAIN

HISTORY

Name: DOB: 12/28/1954 Female
Exam Date: 9/8/14
Referring Phys.: GunnarHeuser,M.D.

This is a 59-year-old female with exposure to mold and mercury. The patient has symptoms of seizures, memory loss, and numbness in hands and left arm.

PROCEDURE

Using a 3 Tesla Siemens Verio MRI Open system, the following sequences were obtained:

- | | |
|---------------------------|-------------------------------------|
| 1) Localizer. | 4) DWI axial. |
| 2) T1 3D sagittal MPRAGE. | 5) SWI axial. |
| 3) T2 FLAIR sagittal. | 6) T2 FLAIR axial. 7) T2 TSE axial. |

FINDINGS

No cavum septum pellucidum is seen. No blood products are visualized. There is no diffusion restriction.
There is no focal or global volume loss.
The posterior fossa is normal.

The sella and parasellar regions are normal. The flow voids of the vessels of the skull base are identified and normal.

The mastoid air cells, paranasal sinuses, and orbits are normal.

There are scattered foci of T2/FLAIR hyperintensity within the supratentorial subcortical white matter located in bifrontal lobes.

IMPRESSION

Scattered foci of T2/FLAIR hyperintensity in the supratentorial subcortical white matter representing gliosis. This can be seen in the setting of prior head trauma, chronic migraine headaches, less likely chronic small vessel ischemic disease given distribution.

B

FUNCTIONAL MRI (fMRI) OF THE BRAIN

HISTORY DOB: 12/28/1954 Female
Exam Date: 9/8/14
Referring Phys.: GunnarHeuser, M.D.

This is a 59-year-old female with exposure to mold and mercury, has symptoms of blackout, seizure, memory loss, head trauma to the left side of brain.

PROCEDURE

Using a 3 Tesla Siemens Verio MRI Open bore system, a functional MRI study was performed using the followings equences:

- 1) MPRAGE.
- 2) Resting state network. The EPI sequences for this resting state BOLD scan were then post processed on a 3D workstation and the independent component analysis performed separating out the various networks.
- 3) Arterial spin labeling.
- 4) Tractography and fractional anisotropy.

FINDINGS

There is a markedly abnormal default mode network with hyperconnectivity of the anterior component of the default mode. This can be seen in the setting of OCD, chronicpain, post-traumatic brain injury and/ordrug abuse.

The fractional anisotropy is low within the body of the corpus callosum with a minimal value of 0.22 in the anterior body .Diffusion tensor imaging is within normal.

The arterial spin labeling demonstrates symmetric diminished flow within bifrontal lobes.

IMPRESSION

1. Abnormal functional MRI with hyperconnected anterior component of the default mode network in the medial orbitofrontal area. This can be seen in the setting of traumatic brain injury, chronic pain, substance abuse and/or OCD.
2. Decreased fractional anisotropy within the body of the corpus callosum with a minimal value of 0.22 in th eanterior body. This is a sign of disorganization of fibers which can be seen in the setting of traumatic brain injury.
3. There is diminished flow within bifrontal regions which is consistent with hypometabolic activity. This can be seen in the setting of traumatic brain injury.